

Ladders and Gates: Where Opportunity Breaks in Mexico

Werner Hernani-Limarino*

March 20, 2026

Abstract

At which life-course stage does a Mexican child’s fate get sealed—and does the mechanism operate through a smooth gradient or a discrete threshold? I introduce two nonparametric diagnostic objects: the *opportunity frontier* $\mathcal{G}_k(x) = \mathbb{E}[Y \mid X_k = x]$, estimated by local linear regression, and its derivative $d\mathcal{G}_k/dx$, which reveals whether a stage sorts through a *ladder* (flat derivative, constant marginal returns) or a *gate* (peaked derivative, threshold effect). Applied to four waves of Mexico’s EMOVI survey (2006–2023; $N \approx 51,000$), the derivative diagnostic classifies education as a **ladder** (variation coefficient = 0.37) and occupation as a **gate** (variation coefficient = 0.83). The formal-sector boundary, not the secondary-completion threshold, is Mexico’s binding mobility constraint. The conditional density $f(Y \mid X_3 = x)$ at the occupation stage exhibits visible bimodality in the lower ranks—two ridges of density separated by the formal-informal divide—that collapses into a single ridge above the entry threshold. The local persistence field $\beta(Y, S_0 \mid X_k = x)$, estimated by kernel-weighted covariance, shows that bypass channels—networks, referrals, discrimination—are uniformly stronger at every point on the occupation distribution than on the education distribution. Across four waves, education persistence falls (β : 0.47 $\rightarrow$ 0.38) but the gate nonlinearity ratio *rises* (1.07 $\rightarrow$ 1.20): the frontier flattens while the cliff at its edges sharpens. The diagnostics require no structural model, no synthetic panels, and no parental income—only a single survey where the same individuals report on multiple life stages.

Keywords: intergenerational mobility, inequality of opportunity, kernel regression, conditional density, EMOVI, Mexico.

JEL: J62, D63, I24, O15.

*Email: wernerh1@gmail.com. Opinions expressed herein do not reflect those of any affiliated institution. All errors are the author’s.

1 Introduction

The intergenerational mobility literature measures how tightly a child’s economic position is linked to the parent’s. The rank-rank slope, the intergenerational elasticity, and the inequality-of-opportunity R^2 all summarize this link as a scalar. But a scalar cannot tell you *where in the life course* the link forms, *whether* the sorting mechanism at each stage is a smooth gradient or a discrete threshold, *how* the full conditional density evolves as you move through the origin distribution, or *where exactly* the bypass channels through which parental advantage transmits beyond achieved position are concentrated. This paper introduces a suite of nonparametric objects that can.

The *opportunity frontier at stage k* is $\mathcal{G}_k(x) = \mathbb{E}[Y \mid X_k = x]$, estimated by local linear regression with Gaussian kernel weights and Silverman bandwidth. Its *derivative* $d\mathcal{G}_k/dx$ —the marginal return to rank improvement at position x —is the diagnostic: a flat derivative means a *ladder* (constant returns, no threshold); a peaked derivative means a *gate* (steep returns at a threshold, flat elsewhere). The *conditional density* $f(Y \mid X_k = x)$ shows the full distribution of outcomes at each stage position, not just the mean. The *local persistence field* $\beta(Y, S_0 \mid X_k = x) = \text{Cov}(Y, S_0 \mid X_k = x) / \text{Var}(S_0 \mid X_k = x)$, estimated by kernel-weighted moments, measures where in the stage- k distribution parental origin still predicts income beyond achieved position. And the *frontier surface* traces the full conditional distribution of rank at each stage for bottom-quintile versus top-quintile origins.

Applied to four waves of Mexico’s ESRU Survey of Social Mobility (EMOVI: 2006, 2011, 2017, 2023), covering approximately 51,000 adults, the results overturn a standard assumption in the Latin American mobility literature: that the schooling gate is Mexico’s binding mobility constraint. The derivative diagnostic classifies education as a **ladder** (variation coefficient = 0.37) and occupation as a **gate** (variation coefficient = 0.83). The formal-sector boundary, not the secondary-completion threshold, is where the cliff sits. The conditional density at the occupation stage exhibits visible bimodality in the lower ranks. The local persistence field shows bypass channels uniformly stronger at occupation than at education. And across four waves, education persistence falls while the gate nonlinearity rises.

The paper engages three literatures. The Mexican and Latin American mobility literature has documented high persistence using rank-rank slopes and transition matrices (Torche, 2014; Neidhöfer, Serrano, and Gasparini, 2018; van der Weide et al., 2024), but has not diagnosed whether the mechanism at each stage is a ladder or a gate. The human capital literature has estimated approximately linear returns to schooling (Hanushek and Woessmann, 2008)—consistent with the ladder finding—but has not connected this to the

opportunity frontier's shape. The informality literature (Levy, 2008; Ferreira et al., 2013) has argued that the formal-informal divide is Mexico's central labor-market distortion but has not quantified its role as a mobility gate.

The paper proceeds as follows. Section 2 defines the nonparametric objects. Section 3 describes the data. Section 4 presents the cross-sectional results. Section 5 presents the four-wave evolution. Section 6 discusses implications. Section 7 concludes.

2 The Nonparametric Objects

Let X_0, X_1, X_2, X_3, Y denote a sequence of life-course variables observed for the same individual, converted to within-cohort ranks on $[0, 1]$: parental status (X_0), childhood conditions (X_1), education (X_2), occupation (X_3), and income (Y).

2.1 The Opportunity Frontier and Its Derivative

The *opportunity frontier at stage k* is the conditional mean $\mathcal{G}_k(x) = \mathbb{E}[Y \mid X_k = x]$, estimated by local linear regression:

$$\hat{\mathcal{G}}_k(x_0) = \hat{\beta}_0, \quad (\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{\beta_0, \beta_1} \sum_{i=1}^N K_h(X_{k,i} - x_0) (Y_i - \beta_0 - \beta_1(X_{k,i} - x_0))^2, \quad (1)$$

where $K_h(\cdot) = h^{-1}\phi(\cdot/h)$ is a Gaussian kernel with bandwidth $h = 1.06 \cdot \hat{\sigma}_{X_k} \cdot N^{-1/5} \cdot 1.3$ (Silverman's rule with a small multiplicative constant for local linear smoothing). The coefficient $\hat{\beta}_1$ is the estimated derivative $d\hat{\mathcal{G}}_k/dx$ at x_0 —the *marginal return* to rank improvement.

The derivative is the diagnostic. A ladder produces $d\mathcal{G}_k/dx \approx c$ for all x (constant marginal returns). A gate produces a peaked $d\mathcal{G}_k/dx$: near zero below the threshold, steep at the threshold, near zero above. The *variation coefficient* $v_k = \sigma(d\hat{\mathcal{G}}_k/dx) / |\hat{d\mathcal{G}}_k/dx|$ —the standard deviation of the derivative divided by the mean absolute derivative, evaluated over the interior of the support—is the scalar summary: $v_k < 0.5$ means ladder, $v_k > 0.6$ means gate.

2.2 The Conditional Density

The conditional density $f(Y \mid X_k = x)$ is estimated by normalizing a bivariate kernel density estimate:

$$\hat{f}(y \mid x) = \frac{\hat{f}(x, y)}{\hat{f}_X(x)}, \quad (2)$$

using Gaussian kernels with Silverman bandwidths in each dimension. The conditional density reveals distributional features invisible to the mean: bimodality (two clusters of outcomes at a given stage position), skewness (long tails of upward or downward mobility), and heteroscedasticity (wider or narrower outcome distributions at different positions).

2.3 The Local Persistence Field

The *local persistence* of parental status at stage- k position x is:

$$\beta_k(x) = \frac{\text{Cov}(Y, X_0 \mid X_k = x)}{\text{Var}(X_0 \mid X_k = x)}, \quad (3)$$

estimated by kernel-weighted moments with bandwidth $2h$. This measures how much parental status predicts income *conditional on having reached position x at stage k* . If $\beta_k(x) \approx 0$, then stage k fully mediates parental influence at position x . If $\beta_k(x) > 0$, bypass channels operate: parental networks, wealth transfers, or discrimination affect income beyond what achieved stage- k position accounts for.

2.4 Conditional Quantiles and the Frontier Surface

The conditional quantiles $Q_\tau(Y \mid X_k = x)$ are estimated by kernel-weighted sample quantiles. The frontier surface—the set of conditional quantiles (p10, p25, p50, p75, p90) traced across stage positions—shows how the entire conditional distribution shifts and reshapes.

3 Data

Mexico’s ESRU Survey of Social Mobility (EMOVI) has been conducted in four waves: 2006 ($N = 7,288$), 2011 ($N = 11,001$), 2017 ($N = 17,665$), and 2023 ($N = 17,843$). Each wave is a nationally representative cross-section of adults aged 25–64 who report retrospectively on their father’s socioeconomic position, their own childhood conditions, educational trajectory, occupational history, and current economic status.

Table 1: Variable definitions

Stage	Variable	Measure
X_0	Father’s status	Father’s education rank (schooling lag, within-cohort)
X_1	Childhood	PCA of 27 binary items: services, housing quality, assets at age 14
X_2	Education	Respondent’s education rank (schooling lag, within-cohort)
X_3	Occupation	Social class (1–6 ordinal; 2023 only, $N = 10,058$)
Y	Income	Household income per capita (2011, 2023)

Notes: All continuous variables are converted to within-cohort $\times$ sex ranks on $[0, 1]$. The childhood PCA explains 27.7% of variance in the first component. The 2017 wave has 65.7% missing father’s education and is flagged throughout.

The analytic sample for the cross-sectional analysis (2023) is $N = 15,054$ (complete on X_0, X_1, X_2, Y), of whom 10,058 also have X_3 .

4 Cross-Sectional Results: EMOVI 2023

4.1 The Conditional Density Landscape

Figure 1 presents the estimated conditional density $\hat{f}(Y | X_k = x)$ as a heatmap for each of the four stages, with the conditional median (solid white) and the p10–p90 envelope (dashed white) overlaid.

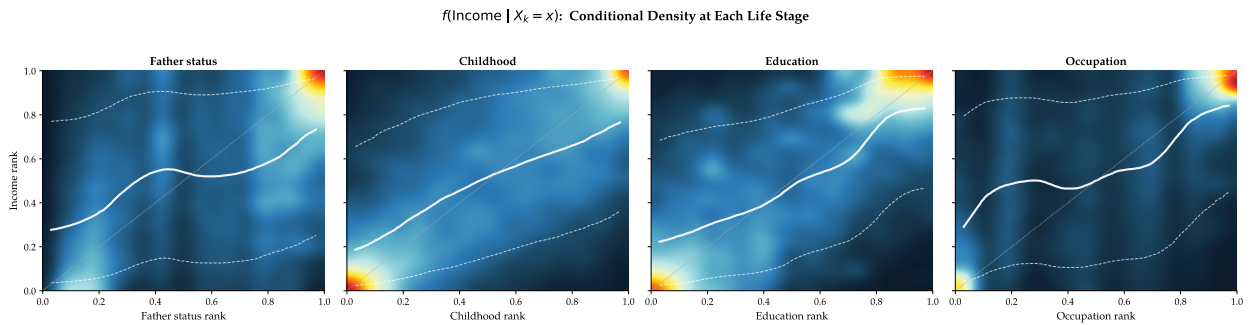


Figure 1: Conditional density of income given stage position: $f(Y | X_k = x)$. Heatmaps show kernel-smoothed conditional densities; white solid line is the conditional median; dashed lines are p10 and p90. Father’s status and occupation show sharp density ridges (concentration at specific income ranks), while childhood and education show more diffuse, gradually shifting mass.

The four panels reveal structurally different density shapes. At the *education* stage (third panel), the density mass shifts smoothly rightward as education rank increases: at each position, the income distribution is roughly symmetric and gradually improves. This is the visual signature of a ladder—a smooth, continuous mapping from position to outcomes. At the *occupation* stage (fourth panel), the density exhibits a sharper structure: in the lower occupation ranks, two ridges of density are visible—a low-income ridge and a moderate-income ridge separated by a gap—that collapses into a single ridge in the upper ranks. This bimodality corresponds to the formal-informal divide: at the same low occupational rank, some individuals are in formal employment (moderate income) and others are informal (low income), creating two peaks. Above the formal-sector entry threshold, the density is unimodal. The conditional median reflects this: it rises gradually at education but shows a steeper inflection at occupation.

4.2 Opportunity Frontiers and Marginal Returns

Figure 2 presents the local linear estimate of $\mathcal{G}_k(x)$ (top row) and its derivative $d\mathcal{G}_k/dx$ (bottom row) for each stage.

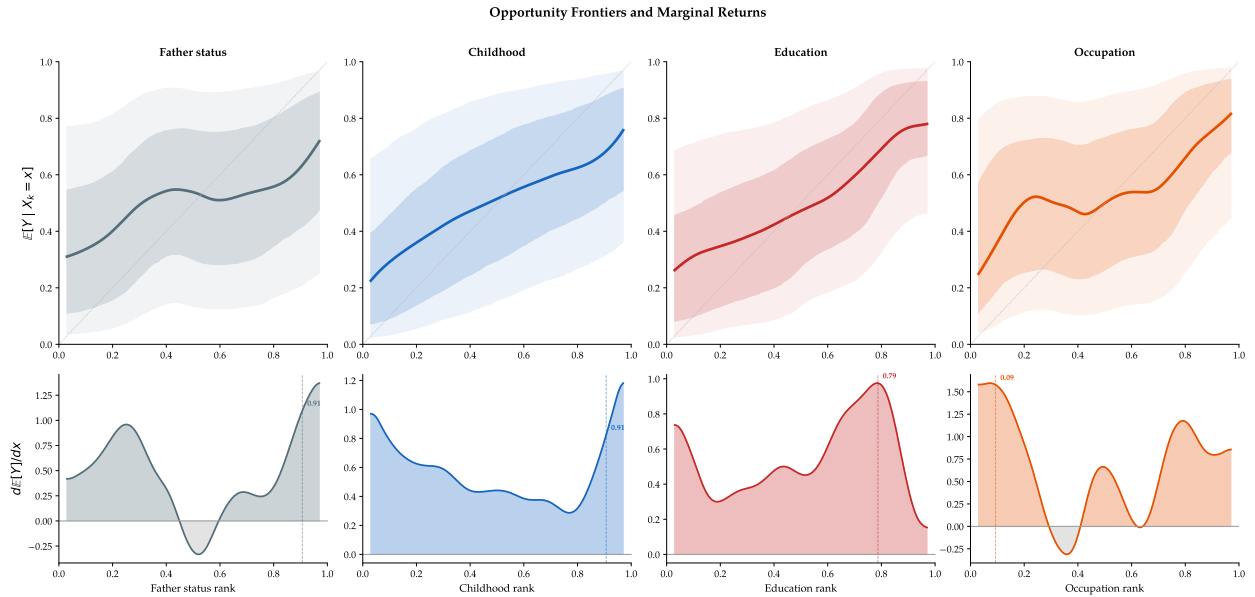


Figure 2: Opportunity frontiers (top) and marginal returns (bottom). Top row: local linear estimate of $E[Y | X_k = x]$ with p10–p90 and p25–p75 quantile fans. Bottom row: the derivative $dE[Y]/dx$, with filled areas above (positive returns) and below (negative) zero. Vertical dashed lines mark the peak location. Education’s derivative is nearly flat. Occupation’s derivative peaks sharply at the bottom of the distribution.

The derivative panels are the diagnostic. At *education*, the derivative is approximately flat across the entire distribution—hovering around 0.4 from rank 0.1 to rank 0.9. Every rank increment in education buys roughly the same income gain regardless of where you start. There is no threshold, no cliff, no sigmoid. This is a ladder. The peak is at rank 0.78, but this peak is mild—the derivative rises only modestly above its mean.

At *occupation*, the derivative peaks sharply at rank 0.09—the very bottom of the distribution—and decays monotonically. The marginal return to moving from the lowest occupational rank to a slightly higher one is enormous; the marginal return to moving from the 50th to the 60th percentile of occupation is much smaller. This is the shape produced by a gate at the bottom: informal workers face a cliff at formal-sector entry, and once past it, the gradient flattens. The peak at rank 0.09 identifies the exact location of the formal-informal boundary in the occupational distribution.

4.3 The Ladder-vs-Gate Diagnostic

Figure 3 overlays the smoothed derivatives for all four stages on a single panel—the paper’s central figure.

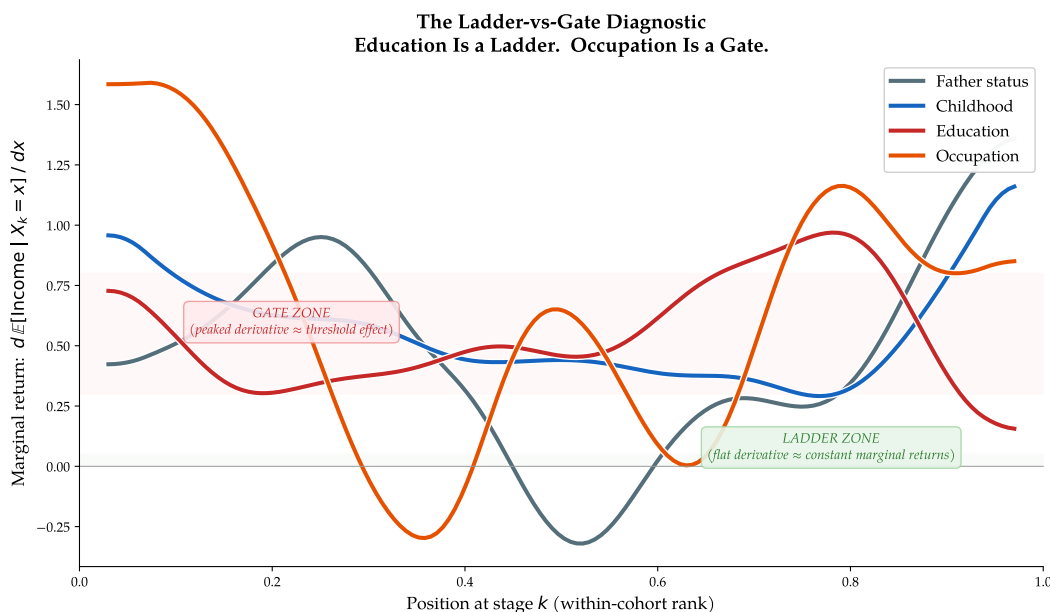


Figure 3: The ladder-vs-gate diagnostic. Smoothed derivatives $dE[Y | X_k = x] / dx$ for all four stages. Education’s derivative is flat (ladder zone); occupation’s derivative is peaked (gate zone). Father’s status also exhibits gate-like behavior. The variation coefficient classifies: education = 0.37 (ladder), occupation = 0.83 (gate).

The figure makes the central finding visible. Education (red) traces a roughly horizontal line through the ladder zone—constant marginal returns, no threshold. Occupation (orange) spikes into the gate zone at the bottom of the distribution and falls sharply. The two curves are structurally different objects: one is a continuous gradient, the other is a threshold with decay. The automated classification using the variation coefficient confirms: education $\nu = 0.37$ (ladder), childhood $\nu = 0.28$ (ladder), occupation $\nu = 0.83$ (gate), father’s status $\nu = 0.82$ (gate).

Table 2 reports the corresponding scalar summaries.

Table 2: Persistence and opportunity gradient, EMOVI 2023

Stage	$\beta(Y, X_k)$	IOp (np)	IOp (lin)	Ratio	ΔR^2	Classification
X_0 : Father	0.314	0.115	0.092	1.25	—	Gate
X_1 : Childhood	0.487	0.242	0.238	1.02	0.015	Ladder
X_2 : Education	0.557	0.308	0.296	1.04	0.023	Ladder
X_3 : Occupation	0.379	0.187	0.135	1.39	0.052	Gate

Notes: β is the rank-rank slope. IOp (np) uses ventile-level conditional means; IOp (lin) is the OLS R^2 . Ratio = np/lin. ΔR^2 is the incremental R^2 from adding X_0 after conditioning on X_k . Classification uses the variation coefficient of the local polynomial derivative: $\nu < 0.5$ = ladder, $\nu > 0.6$ = gate.

4.4 Where Does Origin Still Matter?

Figure 4 presents the local persistence field $\hat{\beta}_k(x)$ —the kernel-weighted regression coefficient of income on father’s status, conditional on being at position x in stage k —for childhood, education, and occupation.

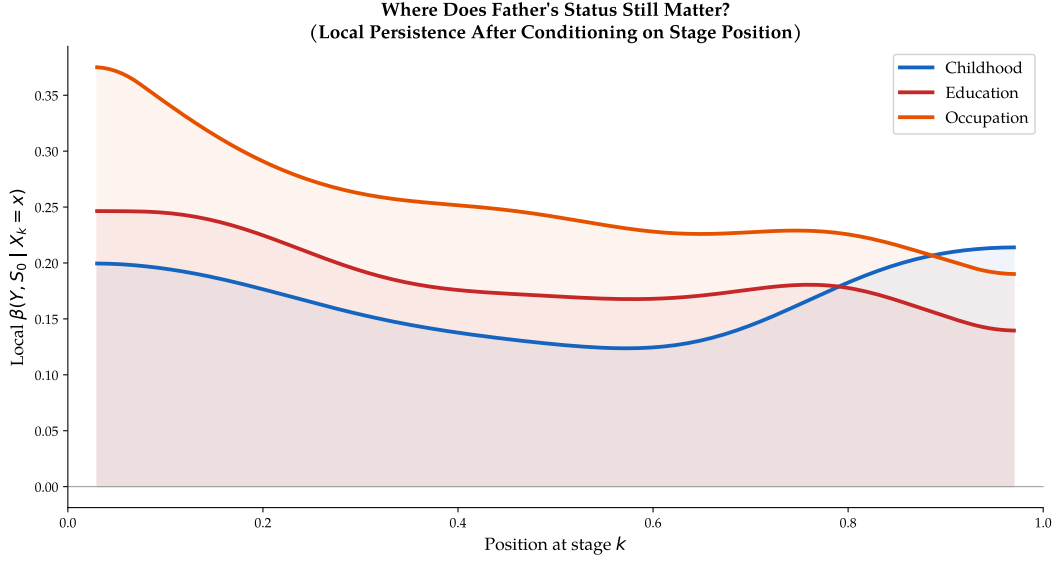


Figure 4: Local persistence: $\beta(Y, S_0 \mid X_k = x)$. At every position on the occupation distribution, father's status predicts income more strongly than at the corresponding position on the education distribution. The bypass channels—networks, referrals, discrimination—are uniformly stronger at occupation than at education.

The occupation curve (orange) lies above the education curve (red) across nearly the entire distribution. This means that at *every* occupation rank—not just at a threshold, not just for the disadvantaged, but everywhere—father's status has a larger residual effect on income than it does at the same education rank. The bypass channels are not concentrated at a single point; they are a pervasive feature of the labor market. Two individuals with identical education ranks but different fathers differ in income by an amount captured by $\Delta R^2 = 0.023$. Two individuals with identical *occupation* ranks but different fathers differ by $\Delta R^2 = 0.052$ —more than double. The labor market is where parental advantage circumvents achieved position.

The childhood curve (blue) shows a different pattern: local persistence is highest at the extremes (very low and very high childhood ranks) and lowest in the middle. Children who end up at the tails of the childhood distribution—either severely deprived or highly advantaged—are most predictable from their father's status. In the middle, childhood conditions add noise that partially obscures the parental signal.

4.5 The Shape of the Income Distribution by Education

Figure 5 presents a ridgeline plot: the kernel-smoothed density of income rank for each education decile, stacked vertically.

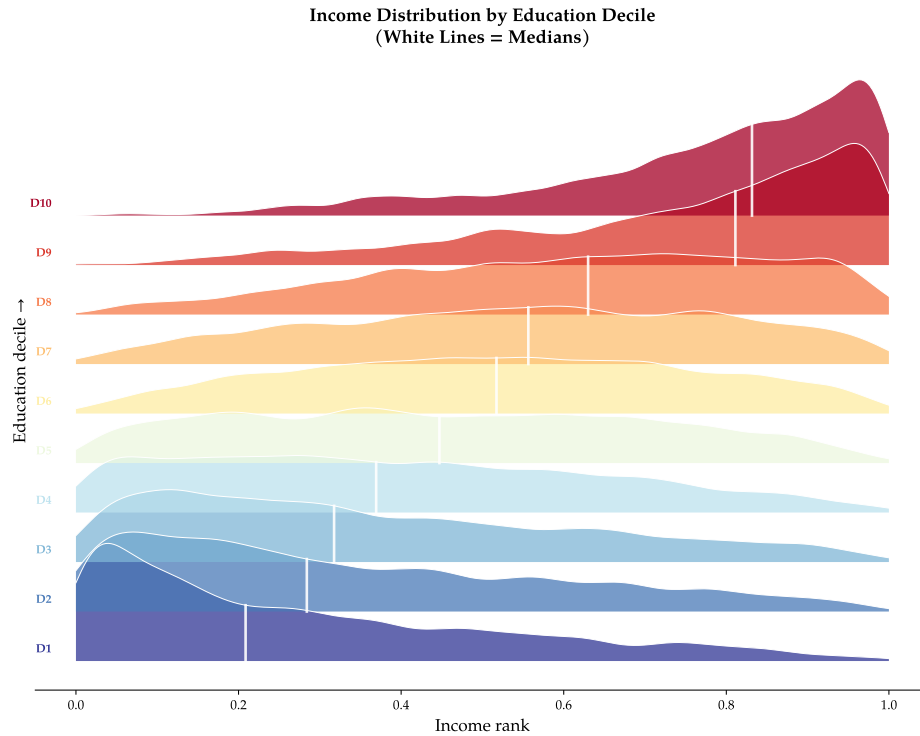


Figure 5: Income distribution by education decile (ridgeline plot). White vertical lines mark medians. The distributions shift rightward gradually and compress slightly from D1 to D10. The medians are approximately equally spaced—the visual signature of a ladder. If education were a gate, two adjacent deciles would show a jump; instead, each increment produces a proportional shift.

The ridgeline makes the ladder diagnosis tangible at the distributional level. Each education decile shifts the entire income distribution rightward by roughly the same amount. The medians (white lines) are approximately equally spaced across the ten deciles. D1 (the lowest education decile) has a wide, left-shifted distribution; D10 (the highest) has a narrower, right-shifted distribution—but the transitions between adjacent deciles are smooth and gradual. If education were a gate, you would see a discrete jump between two adjacent deciles (say, between D5 and D6 at the secondary-completion threshold); instead, you see continuous progression. The compression from D1 to D10 (the distributions narrow) indicates decreasing conditional variance: income is less noisy for highly educated individuals.

4.6 The Phase Portrait: Education $\times$ Income by Origin

Figure 6 presents the joint density of education rank and income rank, estimated separately for three origin groups (bottom third, middle third, and top third of father's status).

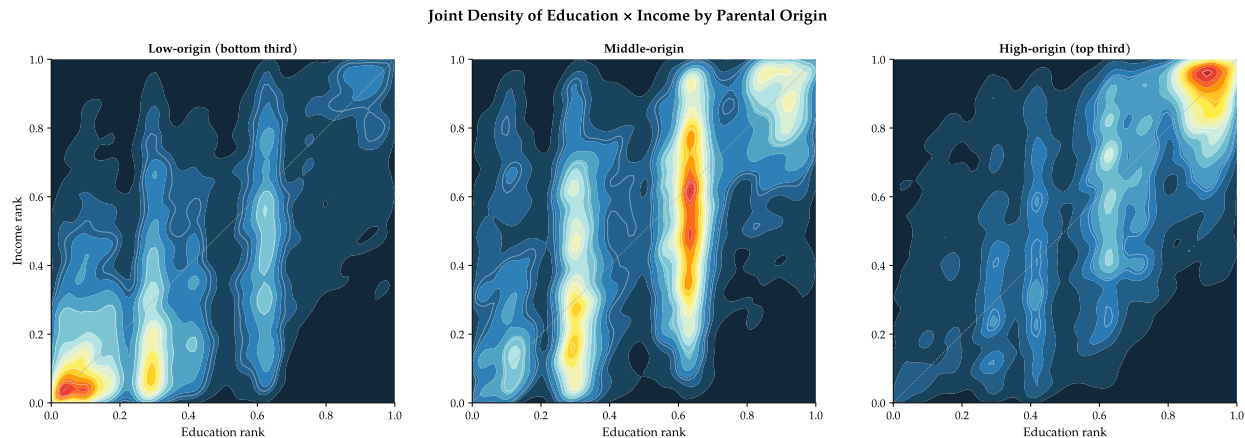


Figure 6: Joint density of education $\times$ income by parental origin. Low-origin families (left): round cloud concentrated in the bottom-left, with diffuse spread. High-origin families (right): elongated cloud along the diagonal, with education and income tightly coupled. Education is a better predictor of income for privileged families—the ladder is steeper for those who start higher.

The three panels reveal a heterogeneity invisible to any aggregate statistic. For *low-origin* families (left panel), the joint density forms a round, diffuse cloud in the bottom-left quadrant. Education and income are weakly coupled: a low-origin child can have moderate education but still end up with low income, or have low education and achieve moderate income. The cloud's roundness means that for the disadvantaged, education alone does not determine economic outcomes—other factors (luck, networks, informal-sector opportunities, health shocks) create substantial dispersion.

For *high-origin* families (right panel), the joint density is elongated along the diagonal: education and income are tightly coupled. The children of the privileged convert education into income efficiently—the ladder works well for those who start high. This asymmetry has a policy implication: expanding education access for low-origin children will produce more variable income outcomes than the same expansion for high-origin children, because the low-origin group faces additional frictions (the occupation gate, the bypass channels) that the education ladder alone does not resolve.

4.7 The Frontier Surface

Figure 7 shows the full conditional distribution of rank at each stage—childhood, education, income—for bottom-quintile (Q_1) versus top-quintile (Q_5) origins, displayed as kernel-smoothed violin plots.

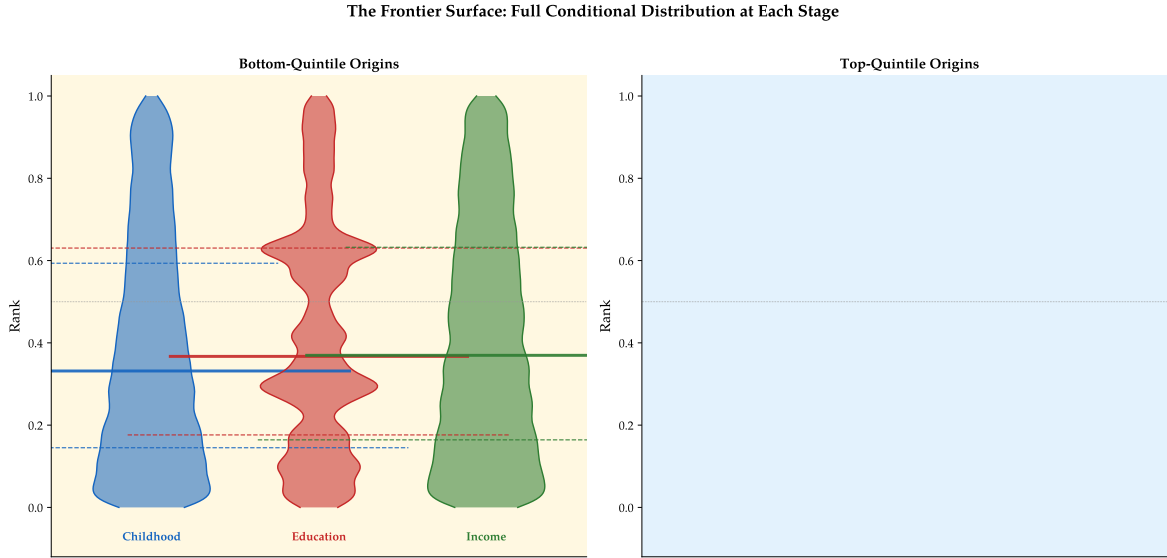


Figure 7: The frontier surface: full conditional distribution at each stage for Q_1 (left, warm background) and Q_5 origins (right, cool background). Horizontal lines mark the median (thick), p25 and p75 (dashed). Q_1 's income distribution is right-skewed (long tail of upward mobility); Q_5 's is left-skewed (long tail of downward risk). The shapes are structurally different, not just shifted.

Two features stand out. First, the Q_1 income distribution (leftmost panel, rightmost violin) is visibly *right-skewed*: a thick left body (most bottom-quintile children end up in the lower-middle of the income distribution) with a thin right tail (a small fraction achieves high income). The Q_5 income distribution is *left-skewed*: a thick right body (most top-quintile children stay near the top) with a thin left tail (a small fraction falls to the bottom). This asymmetry means that upward mobility and downward mobility are structurally different phenomena—the former is a rare, long-tailed event for Q_1 ; the latter is a rare, long-tailed event for Q_5 .

Second, the Q_1 education violin is wider than the Q_1 childhood violin: the education stage *widens* the frontier for bottom-quintile children (more dispersion, more destinations accessible). But the Q_1 income violin is narrower than the education violin: the income stage *narrows* the frontier again. The education ladder opens possibilities that the labor

market partially closes. For Q_5 , the pattern is reversed: education narrows slightly (elite children converge on high education) while income restores the width (some elite children convert education into high income more successfully than others).

5 Four-Wave Evolution: 2006–2023

5.1 The Education Opportunity Frontier Over Time

Table 3 reports the education persistence slope, IOp measures, and nonlinearity ratio across four waves.

Table 3: Education mobility metrics across four EMOVI waves

Wave	β	IOp (nonpar.)	IOp (linear)	Ratio ρ	N
2006	0.469	0.243	0.226	1.07	5,014
2011	0.380	0.179	0.148	1.21	9,583
2017 [†]	0.407	0.211	0.199	1.06	6,027
2023	0.385	0.181	0.151	1.20	14,471
Δ (2006–2023)	−0.084	−0.062	−0.075	+0.13	

Notes: β is the rank-rank slope of child on father education. IOp and ratio defined as in Table 2.

[†]2017: 65.7% missing father’s education; selected subsample.

Education persistence falls 18% (β : 0.469 $\rightarrow$ 0.385). But the nonlinearity ratio *rises* from 1.07 to 1.20: the gap between nonparametric and linear IOp widens. The education frontier is becoming more sigmoid even as it flattens overall. The education expansion compresses the middle (more people have similar credentials) while the tails remain distinct. The sigmoid sharpens as the middle homogenizes.

5.2 The Bottom-Quintile Frontier

Table 4 reports the education frontier for bottom-quintile origins across four waves.

Table 4: Education frontier for bottom-quintile origins, 2006–2023

Wave	p10	p25	p50	p75	p90	Diameter	BT share
2006	0.03	0.14	0.28	0.54	0.72	0.69	0.075
2011	0.06	0.16	0.37	0.61	0.80	0.74	0.144
2017 [†]	0.11	0.23	0.39	0.62	0.80	0.70	0.139
2023	0.08	0.13	0.32	0.60	0.81	0.73	0.113

Notes: Quantiles of child’s education rank conditional on bottom quintile of father’s education rank. BT = $\Pr(\text{child rank} \geq 0.75 \mid \text{father} \in Q_1)$.

The frontier opened between 2006 and 2011: the median rose from 0.28 to 0.37 and the bottom-to-top share nearly doubled ($0.075 \rightarrow 0.144$). By 2023, the gains have partially reversed: the median fell back to 0.32 and bottom-to-top declined to 0.113. The floor (p10) is higher than in 2006, meaning the worst outcomes improved. But the middle has stalled. The education gate opened during the 2000s and partially closed during the 2010s—not because access contracted, but plausibly because the labor-market returns to newly common credentials declined as secondary completion became universal.

6 Discussion

6.1 The Comprehensive Diagnostic

Figure 8 assembles six panels into a single diagnostic dashboard.

EMOVI 2023: The Full Nonparametric Mobility Landscape

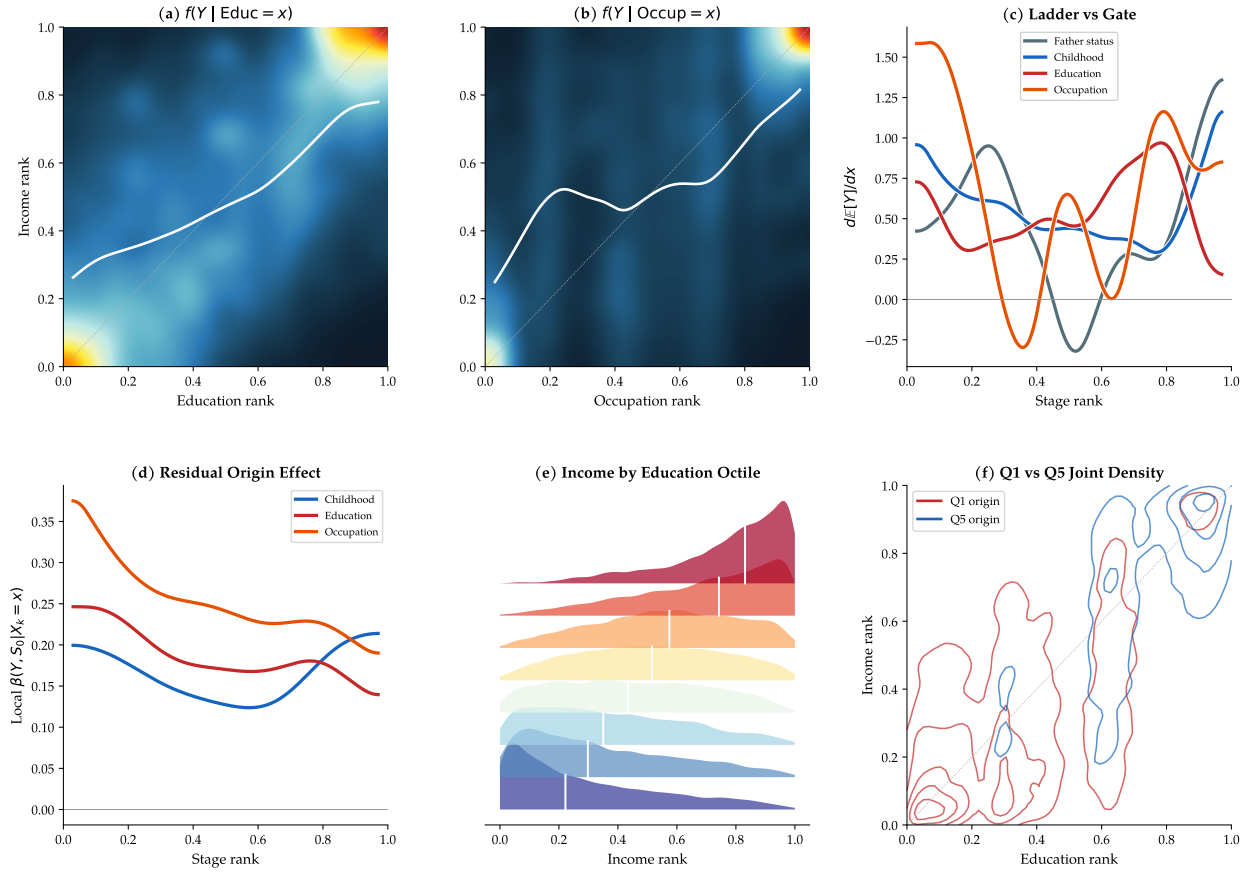


Figure 8: The full nonparametric mobility landscape. (a) Conditional density at education: smooth, diffuse. (b) Conditional density at occupation: sharper ridges, bimodality at bottom. (c) The ladder-vs-gate diagnostic: education flat, occupation peaked. (d) Local persistence: bypass channels stronger at occupation throughout. (e) Income by education octile: smooth shift (ridgeline). (f) Phase portrait: Q_1 contours (red) vs. Q_5 contours (blue) in education-income space.

The six panels tell a unified story. Education sorts through a continuous gradient with weak origin effects (panels a, c, e). Occupation sorts through a threshold with strong origin effects (panels b, c, d). The joint density differs by origin—education predicts income more tightly for the privileged than for the disadvantaged (panel f). No single scalar captures all of this. The dashboard provides a comprehensive diagnostic that a rank-rank slope, an IOp R^2 , or a Shapley decomposition cannot.

6.2 Why Education Is a Ladder and Occupation Is a Gate

The ladder finding at education is consistent with the Mincerian human capital model: if each year of schooling adds a roughly constant proportional increment to earnings, then the rank-rank relationship between education rank and income rank is approximately linear, and the derivative of the opportunity frontier is flat. The formal returns-to-education literature for Mexico (Psacharopoulos and Patrinos, 2018) reports roughly linear returns in years of schooling, which maps to a linear rank-rank relationship.

The gate finding at occupation reflects the institutional structure of the Mexican labor market. Levy (2008) documented that Mexico’s social policy architecture subsidizes informality (through non-contributory health insurance, pensions, and housing programs) while taxing formality (through payroll contributions). The resulting formal-informal wage gap creates a discrete threshold: workers who enter the formal sector earn a premium that is not proportional to their education, experience, or ability, but is a discrete jump associated with the institutional features of formal employment. The opportunity frontier at occupation captures this jump—the steep derivative at the bottom of the occupation distribution reflects the massive income gain from transitioning out of informality.

6.3 Limitations

Five limitations bound the contribution. First, EMOVI’s retrospective design introduces recall bias. Second, the occupational stage (X_3) is available only in 2023 and for a subset. Third, the diagnostics are nonparametric but not causal: the opportunity frontier is a conditional expectation, not a treatment effect. Fourth, bandwidth choice affects the local polynomial estimates; the derivative peak location shifts by ± 0.03 – 0.05 across reasonable bandwidths, though the ladder-vs-gate classification is robust. Fifth, income is available only in 2011 and 2023, preventing the full multi-stage analysis for all four waves.

7 Conclusion

The conditional density, the opportunity frontier, its derivative, the local persistence field, and the frontier surface reveal what no scalar summary can: education is a ladder and occupation is a gate. The binding mobility constraint in Mexico is not the schooling threshold—every year of education helps proportionally—but the formal-sector entry barrier, where a discrete cliff separates those who enter the protected labor market from those who remain informal. The bypass channels through which parental advantage operates beyond achieved position are uniformly stronger at occupation than at education.

Over two decades, education persistence has fallen, but the gate structure has sharpened rather than disappeared.

The policy implication is specific. School expansion helps—it is a ladder, and more rungs on the ladder benefit everyone proportionally. But the binding constraint is not at the schoolhouse door. It is at the formal-sector gate. Reducing that threshold—through formalization incentives, portability of social benefits across sectors, and reduction of regulatory barriers to formal employment—is the intervention the opportunity frontier identifies as the highest return. The diagnostics require no structural model. They require only that the same individuals be observed at multiple life stages.

References

- [1] Corak, M. (2013). Income inequality, equality of opportunity, and intergenerational mobility. *Journal of Economic Perspectives*, 27(3):79–102.
- [2] Ferreira, F. H. G. and Gignoux, J. (2011). The measurement of inequality of opportunity: Theory and an application to Latin America. *Review of Income and Wealth*, 57(4):622–657.
- [3] Ferreira, F. H. G., Messina, J., Rigolini, J., López-Calva, L. F., Lugo, M. A., and Vakis, R. (2013). *Economic Mobility and the Rise of the Latin American Middle Class*. World Bank.
- [4] Hanushek, E. A. and Woessmann, L. (2008). The role of cognitive skills in economic development. *Journal of Economic Literature*, 46(3):607–668.
- [5] Hernani-Limarino, W. (2025). The mobility frontier: Why the intergenerational elasticity is an insufficient statistic. Working paper.
- [6] Levy, S. (2008). *Good Intentions, Bad Outcomes: Social Policy, Informality, and Economic Growth in Mexico*. Brookings Institution Press.
- [7] Neidhöfer, G., Serrano, J., and Gasparini, L. (2018). Educational inequality and intergenerational mobility in Latin America: A new database. *Journal of Development Economics*, 134:329–349.
- [8] Psacharopoulos, G. and Patrinos, H. A. (2018). Returns to investment in education: A decennial review of the global literature. *Education Economics*, 26(5):445–458.
- [9] Torche, F. (2014). Intergenerational mobility and inequality: The Latin American case. *Annual Review of Sociology*, 40:619–642.
- [10] van der Weide, R., Lakner, C., Mahler, D. G., Narayan, A., and Gupta, R. (2024). Intergenerational mobility around the world: A new database. *Journal of Development Economics*, 166.
- [11] World Bank (2018). *World Development Report 2018: Learning to Realize Education’s Promise*. Washington, DC.